Exploratory Data Analysis Project Report

Project: E-Commerce Sales & Customer Behavior Analysis
Dataset: UCI Online Retail

Project: EDA	**Language:** Python (Pandas, Matplotlib, Seaborn)
Dataset Size: 541,909 rows | 8 columns	**Cleaned Rows:** 522,710 valid purchase rows

1. Project Goal

The goal of this project is to apply Exploratory Data Analysis (EDA) on an e-commerce dataset using Python. I worked with the UCI Online Retail dataset, which contains over 540,000 real sales records. My objective was to clean the data, find errors or missing values, create charts to answer business questions, and write down simple, practical recommendations based on what the data shows.

2. Understanding the Raw Data & Data Cleaning

When I loaded the dataset into Pandas, I checked for missing values, duplicates, and weird numbers. Here is what I discovered:

- **Missing Values:** The dataset has 8 columns. The official website documentation said there were no missing values, but df.isnull().sum() showed that CustomerID had 135,080 missing rows (around 25%) and Description had 1,454 missing rows.
- **Duplicates:** There were 5,268 duplicate rows. I dropped them using df.drop_duplicates().
- **Cancelled Orders:** Invoices starting with 'C' mean the customer cancelled the order. There were 9,288 cancelled invoices and 10,624 rows where Quantity was zero or negative.
- **Wrong Prices:** There were 2,517 rows where UnitPrice was zero or negative (such as system error codes or write-offs like -11,062.06).
- **Non-Product Codes:** I noticed codes like POST (postage), D (discount), and M (manual) were stored in StockCode, and some descriptions had warehouse notes like 'damaged' or 'check'.

How I Cleaned the Data

- **Step 1:** Removed the 5,268 duplicate rows.
- **Step 2:** Separated all cancelled rows into a new dataframe called df_cancelled (10,587 rows) so they wouldn't mess up the sales numbers.
- **Step 3:** Kept only valid purchases where Quantity > 0 and UnitPrice > 0. This gave a clean dataframe df_clean with 522,710 rows.
- **Step 4:** Created new helper columns: TotalPrice (Quantity * UnitPrice), Year, Month, YearMonth, DayOfWeek, and Hour.

Summary of Numbers Before and After Cleaning

Column	Clean Min	Clean Median	Clean 99th Pct	Clean Max
Quantity (units)	1	4 units	100 units	80,995 units
UnitPrice (£)	£0.04	£2.08	£16.63	£11,062.06
TotalPrice (£)	£0.06	£9.90	£179.00	£168,469.60

3. Visualizations and What I Learned (EDA Findings)

1. Most Products are Cheap Gifts Bought in Packs (Histograms)

From plotting histograms, I noticed the median price is only £2.08, and the median spend per item line is £9.90. Most customers buy affordable gifts. Also, the Quantity chart showed spikes at 6, 12, 24, and 36. This told me that many customers are small shop owners or wholesalers who buy full boxes or packs, not just single items.

2. The Netherlands is a Big Wholesale Buyer (Box Plots)

When comparing countries using box plots, most European countries (like Germany and France) had a median quantity of 10 to 12 items. However, the Netherlands had a huge median of 72 items per line! This proved that customers in the Netherlands are buying in bulk to resell.

3. Popular Products & Busy Store Hours (Bar & Count Charts)

- **Top Revenue Products:** The highest selling product by money made is 'REGENCY CAKESTAND 3 TIER' (£174,156.54), followed by 'PAPER CRAFT , LITTLE BIRDIE' (£168,469.60) and 'WHITE HANGING HEART T-LIGHT HOLDER' (£106,236.72).
- **Busy Hours:** Most orders happen between 10 AM and 3 PM, with peak times at 12:00 PM and 3:00 PM. Almost nobody orders at night.
- **Working Days:** Thursday is the busiest day of the week. There are zero orders on Saturday, which means the business was closed on Saturdays.

4. Sales Shoot Up at the End of the Year (Time Series Line Chart)

The line chart showed monthly sales were steady around £500k to £750k from January to August. Starting in September, sales jumped up quickly and hit a massive peak in November 2011 of £1.45 Million! This happens because people are stocking up on gifts for Christmas and the holiday season.

5. A Few Customers Bring Most of the Money (Pareto Curve)

I grouped sales by CustomerID and plotted a cumulative line. I found that: The top 10% of customers bring in 61.4% of total sales, and the top 20% of customers bring in 74.6% of total sales. This follows the 80/20 rule: a small group of repeat wholesale customers drives most of the business.

6. High Return Rate

Every single month, about 15% of orders end up being cancelled or returned. This is a big operational problem because packaging and handling returns costs the store time and money.

4. Simple Business Recommendations

- **1. Take Care of the Top 20% Customers:** Since the top 20% of customers make up almost 75% of sales, the store should give them special discounts, fast shipping, and personal support so they don't leave.
- **2. Stock Up Early for Christmas:** Because November sales double compared to summer, the store must order stock from suppliers in July and August so products don't go out of stock.
- **3. Better Shipping to the Netherlands:** Dutch customers buy huge quantities (72 items median). Setting up a partner warehouse in Europe or cheaper bulk shipping will help grow this market.
- **4. Fix the 15% Cancellation Rate:** Show real-time stock counts on the website. If an item is sold out, don't let customers order it so they don't have to cancel later.
- **5. Staff Shifts for Peak Hours:** Have the warehouse packing team ready from 10 AM to 4 PM on weekdays, especially on Thursdays, when order volume is highest.

5. Conclusion & What I Learned

As a beginner in Exploratory Data Analysis, this project helped me understand that real-world data is messy: it has missing IDs, duplicate rows, negative numbers, and cancelled orders. Cleaning the data first was the most important step. Once the data was clean, using simple charts like histograms, box plots, bar charts, and line charts made it easy to discover clear business patterns like holiday seasonality, wholesale buyer habits, and peak shopping hours.